

AI-Driven and IoT-Integrated Farming Decision Support System

*in partial fulfillment of the requirements for the award of the degree
of Bachelor of Technology in*

COMPUTER SCIENCE & ENGINEERING
(DATA SCIENCE)

Submitted by

PREKSHITA MOTGHARE
STEVEN TITUS
VIDHI CHAMAT

Under the guidance of

Dr. TEJAL IRKHEDE

Academic Year 2024-25

**Department of Computer Science & Engineering
(Data Science)**



ARISE & SHINE

**ST. VINCENT PALLOTTI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

Wardha Road, Gavsi Manapur, Nagpur

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CERTIFICATE

Certified that this project report “**AI-Driven and IoT-Integrated Decision Support system**” is the bonafide work of “PREKSHITA MOTGHARE, STEVEN TITUS, VIDHI CHAMAT” who carried out the project work under my supervision in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science & Engineering (Data Science)**

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ABSTRACT

Agriculture remains a critical pillar of economic stability and food security, particularly in developing nations. However, the sector is increasingly challenged by climate variability, soil degradation, lack of data-driven decision-making tools, and financial constraints faced by farmers. This project proposes an AI-Driven and IoT-Based Farming Decision Support System (FDSS) to address these challenges through the integration of advanced technologies into traditional agricultural practices.

By leveraging artificial intelligence for data analysis and prediction, and IoT for real-time field monitoring, the FDSS aims to assist farmers in making informed decisions throughout the crop lifecycle. Core features include crop recommendation, yield prediction, pest and disease detection through image analysis, smart irrigation, customized bio-fertilizer formulation, and basic financial planning tools. Designed to be cost-effective, scalable, and accessible on mobile devices, this platform targets small to medium-scale farmers with limited technical knowledge.

The project envisions transforming conventional farming into a smart, data-driven, and sustainable practice, ultimately enhancing productivity, increasing profitability, and contributing to climate-resilient agriculture

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ii. LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURES

- **NPK:** Nitrogen, Phosphorus, Potassium (Soil nutrients)
- **ARIMA:** Auto Regressive Integrated Moving Average (Time-series forecasting model).
- **CNN:** Convolutional Neural Network (Deep learning model for image classification)
- **IoT:** Internet of Things (Network of devices connected to the internet for real-time data collection)



CHAPTER 1 INTRODUCTION

1.1 INTRODUCTION

Agriculture is the foundation of many countries' economies, especially in developing nations where a large part of the population depends on farming for their livelihood. It plays a key role in providing food, employment, and income to millions of people. In rural areas, agriculture is not just an occupation-it is deeply connected to the culture, society, and way of life. It also contributes significantly to a country's Gross Domestic Product (GDP) and food security. However, despite its importance, the agricultural sector faces many serious challenges today.

One of the biggest problems is climate change. The weather has become unpredictable, and farmers are often unsure about rainfall, temperature, or the best time to plant and harvest crops. Extreme weather events like floods and droughts have become more frequent, damaging crops and reducing yields. Another issue is the declining quality of soil, caused by years of overusing chemical fertilizers and pesticides. This leads to poor plant growth and lower productivity.

Many farmers also lack access to proper information, modern tools, and expert advice. They don't always know which crop is best for their soil or how to deal with pests and diseases effectively. In addition, they face difficulties with managing water use, buying expensive farm inputs, and getting fair prices in the market. All of this makes farming a risky and often unprofitable job. As a result, many young people are leaving villages in search of better job opportunities in cities, which is affecting both agriculture and rural development.

To solve these problems and improve farming, new technologies like Artificial Intelligence (AI) and the Internet of Things (IoT) are being used. These technologies are helping to create what is known as "smart farming" or "precision agriculture."

AI uses computer systems that can learn from data and make intelligent decisions. In farming, AI can analyze large amounts of data about soil, weather, crops, and pests to help farmers make better decisions. For example, AI can suggest which crop to plant, predict how much harvest a farmer might get, detect plant diseases early using images, and even forecast future market prices.

On the other hand, IoT involves using devices like sensors, drones, and smart machines that collect real-time information from the field. These sensors can monitor soil moisture, temperature, humidity, and plant health continuously. This real-time data helps farmers understand what is happening on their farm at any moment and take quick action when needed. Together, AI and IoT can help reduce waste, use fewer resources, increase crop yield, and protect the environment. This combination leads to precision farming, where everything from watering to fertilizing is done in the right amount, at the right time.

However, in many developing countries, small and medium-scale farmers still don't have access to such advanced systems due to high costs, low awareness, and lack of infrastructure. That's why there is a strong need for an affordable, simple, and easy-to-use solution that can bring these modern tools to every farmer.

In this project, we propose the AI-Driven and IoT-Based Farming Decision Support System (FDSS). This is an intelligent, end-to-end platform specially designed to help farmers throughout the entire farming process—from crop selection to harvesting and even selling in the market. The system combines AI and IoT technologies in a simple mobile application that can be used even by farmers with little or no technical knowledge.

Key features of this system include:

- Smart crop selection based on soil and climate data.
- Yield prediction to help with planning and budgeting.
- Pest and disease detection through AI-powered image analysis.
- Smart irrigation based on real-time soil moisture levels.
- Custom bio-fertilizer advice based on soil nutrients.
- Basic financial tools like budgeting, market price forecasting, and profit planning.

The system also supports offline functionality, which is important in rural areas with weak internet connectivity. It focuses on making technology accessible to farmers at the grassroots level. By using this system, farmers can improve their productivity, earn better profits, reduce risks, and adopt sustainable practices. More importantly, it empowers rural communities, slows down urban migration, and supports the goal of climate-resilient agriculture. This project aims not only to modernize farming but also to make it a profitable, secure, and dignified profession for future generations.

1.2 LIERATURE REVIEW

Literature Review			
Paper Title	Authors	Limitations	How our proposed project overcomes this
Crop Yield Prediction using Machine Learning Algorithms	Patil, S. K., & Kale, V. R.	Limited dataset scope, making it less generalizable to different regions.	Uses diverse datasets from multiple regions to improve generalizability.
IoT-Enabled Bio-Organic Fertilizer Monitoring for Residual Palm Materials for Sustainable Agriculture	Weena Janratchakool, Parinya Jansengrat	Limited crop scope – Focuses only on palm-based fertilizers.	Supports various crops and provides customized bio-fertilizer recommendations.
IoT and AI-Based Intelligent Agriculture Framework for Crop Prediction	Pushpa Singh, Murari Kumar Singh	Requires consistent internet access, which may not be available in rural areas.	Implements offline processing and local storage to work in low-connectivity areas.

Table 1.2 Research Paper for Literature Review

Smart farming systems are most useful when they are easy to access, reliable, and can work well in rural areas. However, many farmers live in places with poor internet, power issues, and limited technical help. Most modern AI and IoT farming systems need constant internet access to work with cloud platforms. This makes them hard to use in villages where connectivity is weak or expensive. For example, the system in the research paper “IoT and AI-Based Intelligent Agriculture Framework for Crop Prediction” by Pushpa Singh and Murari Kumar Singh works well but depends heavily on the internet, which limits its use in remote areas.

To solve this issue, our AI-Driven and IoT-Based Farming Decision Support System (FDSS) uses a hybrid model that can work both online and offline. The system is built using edge computing, which means that most tasks like data processing, disease detection, and irrigation advice are done directly on small devices such as Raspberry Pi, NodeMCU, or ESP32 - without needing the internet. This allows farmers to get real-time help even when they are offline. We also

use lightweight AI models that run directly on devices like smartphones or tablets using tools such as TensorFlow Lite. These models help in crop suggestions, yield prediction, and disease detection. Since they don't need much memory or processing power, they are perfect for farmers using low-cost phones. To save and manage data when offline, we use local databases like SQLite or TinyDB, which store sensor data, user input, and alerts safely until an internet connection becomes available.

This offline-first approach makes smart farming more affordable, accessible, and dependable, especially for small farmers in remote areas. Research by Rajak et al. (2023) and Yousaf et al. (2023) supports this method, showing that edge-based systems work better in low-resource settings. Offline features also improve usability during disasters or power cuts and allow farmers to continue using the system without interruptions. As a result, FDSS offers a practical, inclusive, and scalable solution that brings the power of AI and IoT to even the most rural farming communities.

1.3 SCOPE OF WORK

The Primary aim of this project is to design and implement comprehensive smart farming decision support system that integrate the capabilities of artificial intelligence (AI) and the Internet of things (IoT). The proposed system is structured into two major modules each addressing critical pain points in modern agricultural. The project not only focuses on increasing productivity and resource efficiency but also emphasize sustainability, accessibility and cost effectiveness for farmers, especially in data-scarce or low-infrastructure regions.

AI-Driven Decision Support System

This module serves as the intelligence core of the Farming Decision Support System (FDSS), designed to empower farmers with timely, data-driven agricultural recommendations using machine learning and deep learning techniques. Its primary objective is to assist in smart agricultural decision-making across various stages—from pre-sowing to post-harvest. The module processes complex agricultural data through AI algorithms and delivers actionable insights that enhance productivity, reduce costs, and promote sustainability.

Crop and Fertilizer Recommendation:

The system utilizes supervised machine learning algorithms, particularly classification models like Decision Trees, Random Forests, and XG Boost, to analyse various environmental and agronomic factors such as soil pH, moisture level, NPK composition, historical crop yields, and climatic conditions. Based on this input, it recommends the most suitable crop varieties for the upcoming season, ensuring optimal yield potential. Additionally, the model suggests appropriate fertilizer types-organic, bio-fertilizers, or chemical- depending on crop requirements and soil health, thereby promoting balanced nutrient application and preventing over-fertilization.

Crop Yield Prediction and Harvest Planning:

To support agricultural planning and inventory management, the system incorporates robust yield forecasting features using time-series models such as ARIMA (Auto Regressive Integrated Moving Average), Prophet, and LSTM (Long Short-Term Memory) networks. These models analyse historical yield records, seasonal trends, and weather forecasts to predict the expected output of a given crop. Accurate yield prediction enables farmers to make informed decisions regarding harvest timing, labour allocation, storage planning, and financial forecasting, thereby reducing waste and post-harvest losses.

Pest and Disease Detection via Image Classification:

Early identification of pests and crop diseases is a major challenge for farmers. This system integrates a Convolutional Neural Network (CNN)-based image classification model trained on large agricultural datasets (such as Plant Village) to detect symptoms from plant leaf images. Farmers can use a smartphone to capture images of affected plants, which the model then processes to diagnose common issues like blight, mildew, rust, or pest damage. Once detected, the system provides treatment recommendations, including organic, chemical, or integrated pest management solutions. This feature minimizes crop damage and reduces the indiscriminate use of pesticides.

Rainfall Forecasting and Smart Irrigation Scheduling:

Accurate rainfall forecasting is essential for efficient water management. By integrating real-time meteorological APIs and analysing historical rainfall patterns, the system applies regression and machine learning models to predict upcoming rainfall events. Based on these forecasts, it recommends optimized irrigation schedules to ensure that crops receive the necessary amount of water without wastage. This approach promotes water conservation-especially in drought-prone areas-and enhances plant health by maintaining optimal soil moisture levels.

Financial Tools and Price Prediction:

Economic decision-making is critical to a farmer's success. This module includes budgeting tools that help estimate operational costs-including seeds, fertilizers, labour, irrigation, and pest control-and compare them with expected yields and market prices. Using regression and time-series forecasting models (such as Facebook Prophet), the system predicts price trends for major crops, helping farmers identify the best time to sell their produce to maximize profits. Additionally, it can suggest alternative market opportunities based on regional price data.

Weather-Based and Eco-Friendly Advisory System:

In addition to crop-specific recommendations, the system provides environmental advisories based on real-time weather conditions. For instance, if rainfall is expected, it may advise delaying pesticide application or harvesting. During dry periods, it might recommend the use of soil cover techniques or modified irrigation strategies. These advisories promote environmentally friendly practices, minimize the adverse impacts of chemical use, and enhance the overall resilience of the farming system.

Genetic Crop Improvement Suggestions:

In the long term, the module also explores the application of Reinforcement Learning (RL) and Genetic Algorithms (GAs) to simulate crop breeding decisions. By analysing gene traits, disease resistance patterns, and environmental adaptability, these models offer theoretical recommendations for hybrid crop development. Although this functionality is currently in the research and development phase, it lays the groundwork for future innovations in intelligent seed selection and crop bioengineering-empowering farmers and researchers to cultivate crops that are well-suited to regional challenges.

IoT-Integrated Bio-Fertilizer Formulation:

This feature focuses on generating customized bio-fertilizer recommendations based on real-time data collected from IoT soil sensors. It evaluates the nutrient content (such as nitrogen, phosphorus, and potassium levels) and other parameters like pH and moisture. The system then formulates eco-friendly, balanced bio-fertilizer suggestions tailored to the specific crop and soil conditions. This promotes healthier soil, reduces dependence on chemical fertilizers, and supports organic farming practices that improve long-term soil fertility and crop performance.

1.4 OPERATING TECHNOLOGIES: HARDWARE AND SOFTWARE TECHNOLOGIES

1. Programming Language:

Primary language for backend logic, data analysis, and AI model building.

Selected due to simplicity, extensive library support, and compatibility with IoT devices and AI environments.

2. Machine Learning Frameworks:

- **Scikit-learn:** Utilized for developing conventional ML models such as Decision Trees, Random Forest, and Regression for crop suggestion, price prediction, and financial planning.
- **TensorFlow/Keras:** Utilized to develop deep learning models such as CNNs (for pest and disease diagnosis) and LSTM (for weather and yield prediction).
- **TensorFlow Lite:** Supports AI model deployment on edge platforms such as Raspberry Pi for offline prediction.

3. Data Handling:

- **Pandas:** Utilized to handle data manipulation, preprocessing, and structured data handling (e.g., yield records, soil data).
- **NumPy:** Used for numerical computation, matrix management, and optimization of machine learning operations.

4. Visualization Libraries:

- **Matplotlib:** For static and interactive plots while model validation and data visualization.
- **Seaborn:** For high-level statistical plots such as heatmaps, distribution plots, and correlation matrices.

5. IoT Platforms:

- **Arduino:** Interfaced with sensors (pH, NPK, temperature, moisture) to acquire field data.
- **Raspberry Pi:** Used as the edge controller and edge computing platform; performs data processing locally and sends it to the cloud.
- **AWS IoT Core:** Handles device connectivity, secure messaging, and real-time cloud communication between the system backend and sensors.

6. Cloud Services:

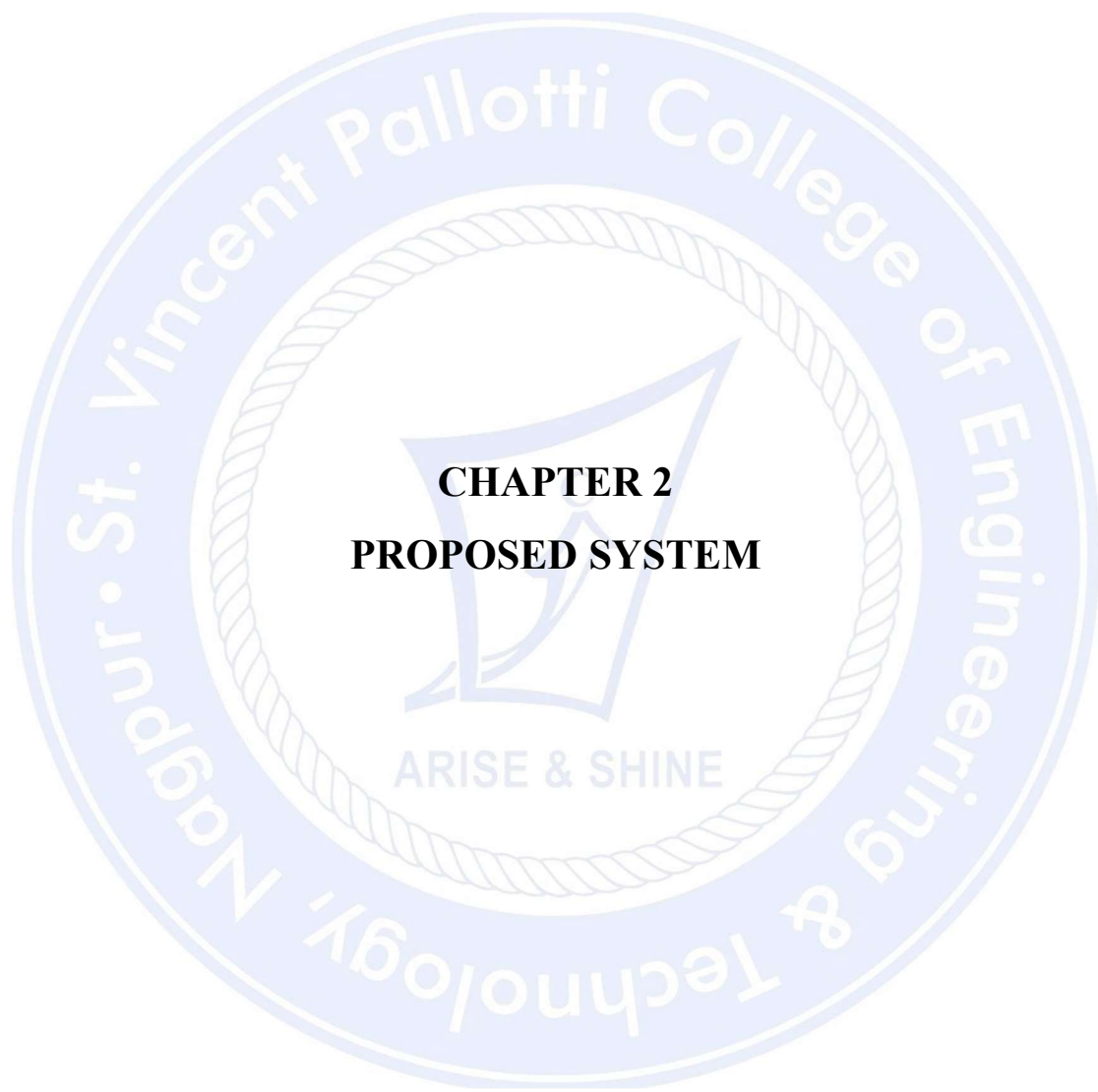
- **AWS S3:** Stores sensor logs, crop images, and historical datasets.
- **AWS Lambda:** Runs serverless functions, including running irrigation alerts or pest control recommendations off real-time inputs.

7. Databases:

- **Firebase:** Offers real-time synchronization for mobile/web apps and user data management.
- **PostgreSQL:** Holds structured and relational data like sensor metadata, user profiles, and financial models; geospatial extensions for future growth.

8. Web Development: React.js, Flask/Django

- **React.js:** Constructs dynamic and responsive user interfaces; modular frontend design and real-time updates.
- **Flask:** Manages lightweight backend services, model API endpoints, and microservices.
- **Django:** Used for full-stack web development, backend management, and secure authentication for administrators and users.



2.1 OBJECTIVES OF PROPOSED SYSTEM

Below are the principal objectives of the AI-Driven and IoT-Integrated Farming Decision Support System (FDSS). They are aimed at addressing some of the top agricultural challenges employing intelligent technologies in making intelligent decisions, effective utilization of resources, and eco-friendly farming.

- **Precision Crop and Fertilizer Recommendation:**

To provide accurate and tailored recommendations for crop selection and fertilizer use based on actual environmental data (soil pH, NPK content, moisture, temperature) and historical trends using AI models. This allows farmers to select the most suitable crops for their region and maintain balanced soil health through effective fertilization, leading to higher productivity and reduced waste.

- **Predictive Yield Forecasting and Harvest Planning:**

To forecast crop yields from the time-series analysis of weather patterns, soil type and historical yield trends with machine learning techniques like LSTM and ARIMA. This can help farmers plan harvest seasons in advance, forecast revenues, and optimize logistics like man power and storage management.

- **Early Pest and Disease Detection:**

To integrate image classification using Convolutional Neural Networks (CNNs) for early crop disease detection and infestation by pests. The farmers would take a photo through a smart phone and obtain instant diagnostic support, enabling timely interventions and reducing crop loss due to delayed detection.

- **Smart Irrigation and Water Management:**

To optimize irrigation schedules through AI-based forecast using weather prediction and IoT-backed real-time measurement of soil moisture. This keeps water use effective, prevents over-irrigation or under-irrigation, and promotes sustainable use of water resources, especially in drought-prone regions.

- **Automated Bio-Fertilizer Formulation:**

To empower farmers with real-time insights into soil nutrient status using IoT sensors and provide personalized bio-fertilizer recommendations using decision trees and optimization algorithms. This objective enables green farming by reducing chemical application and encouraging the use of sustainable, location-tailored organic inputs.

- **Data-Driven Financial and Market Planning:**

To assist farmers in budgeting costs, ascertaining returns, and ascertaining optimal sale times using AI-capable budgeting apps and pricing estimation models. It aids in more informed economic decisions, enhanced financial control, and superior profitability through market trends awareness.

- **Integrated Weather-Based Advisory Services:**

To deliver automated and context-based farm advisories based on real-time and forecasted weather conditions. These include sowing time advisories, pesticide application advisories, and harvesting advisories, which enhance decision-making and crop safety during extreme climate events.

2.2. USER REQUIREMENTS SPECIFICATION

The user requirements for the proposed system are categorized into functional requirements and non-functional requirements:

2.2.1 FUNCTIONAL REQUIREMENTS:

Real-Time Environmental Monitoring:

The system should be able to receive and show real-time data from IoT sensors on significant environmental parameters like soil moisture, temperature, humidity, light intensity, and air quality.

Data-Driven Crop Advice:

Based on AI computation of sensor readings, historical patterns, and local agricultural databases, the system ought to provide recommendations on what crops to plant given current conditions.

Automated Irrigation Control:

The system ought to be able to trigger irrigation systems by pre-determined moisture levels in the soil, optimizing water usage and reducing human involvement. By machine learning software, the system must alert or predict expected infestations or crop illness and inform the user with precautions.

Yield Projection and Farm Performance Reports:

The system should also be in a position to estimate future yield on the crop basis from steady sensor input as well as using AI-based models and present pictorial reports based on performance in a time-keeping manner.

Decision Support Alerts and Notices:

Periodic suggestions and notifications must be dispatched to users through the dashboard and optionally SMS or mobile application alerts to empower farmers to make the most informed choices.

Data Logging and Visualization of Historical Trends:

All the information collected must be stored and be available for inspection. Users must be able to view data trends through interactive graphical aids like graphs and charts.

2.2.2 NON-FUNCTIONAL REQUIREMENTS:

User Interface:

The system should be user-friendly and interactive, supporting easy operation by farmers, agricultural officials, and researchers with various levels of technical skills.

Flexibility and Scalability:

It should be a modular and scalable system that can accommodate various sizes and types of farms, various sensor types, and increasing amounts of data in the long run.

Privacy and Data Security:

The sensitive farm data must be protected with stringent security measures such as encryption of data transmission and role-based access controls.

High Reliability and Performance:

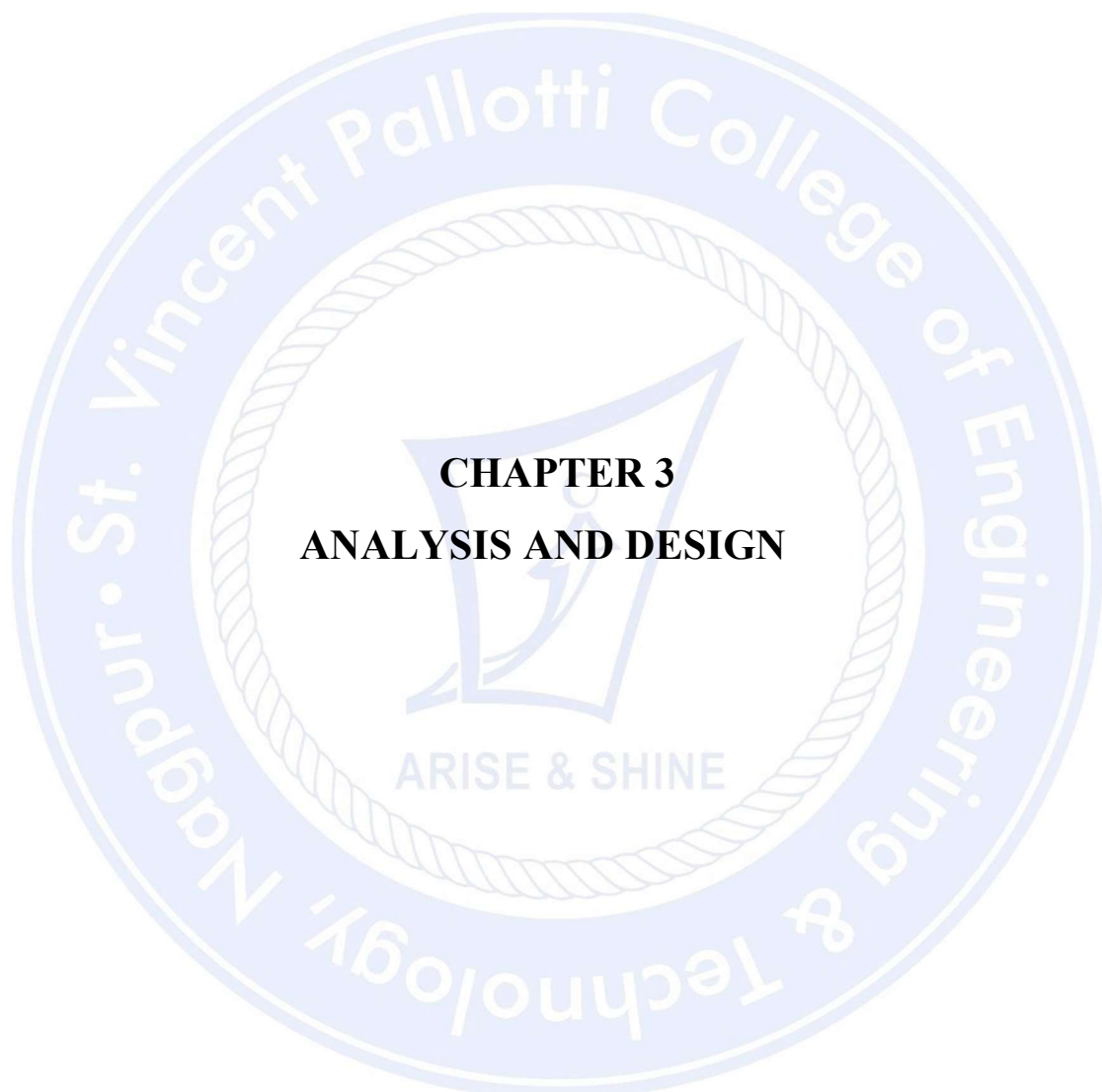
Reliable performance, no downtime, and optimal processing of enormous amounts of data from IoT sensors and AI systems are conditions the system must meet.

Cross-Platform Accessibility:

The platform must work flawlessly on desktops, tablets, and smartphones so that smooth access can be made regardless of device type or geographical location.

Modular Architecture:

A modular architecture must be utilized in order to facilitate future upgrades, i.e., adding new sensors, AI models, or connectivity with outside agricultural databases.



CHAPTER 3

ANALYSIS AND DESIGN

This chapter provided detailed analysis and design of the AI-Driven and IoT-Integrated Farming Decision Support System, covering its structure and functionality. It includes the ER Diagram, Module Specifications, Data Flow Diagrams, Table Specifications, and UI Design to illustrate the system's overall architecture and operations.

3.1. ENTITY-RELATIONSHIP DIAGRAM

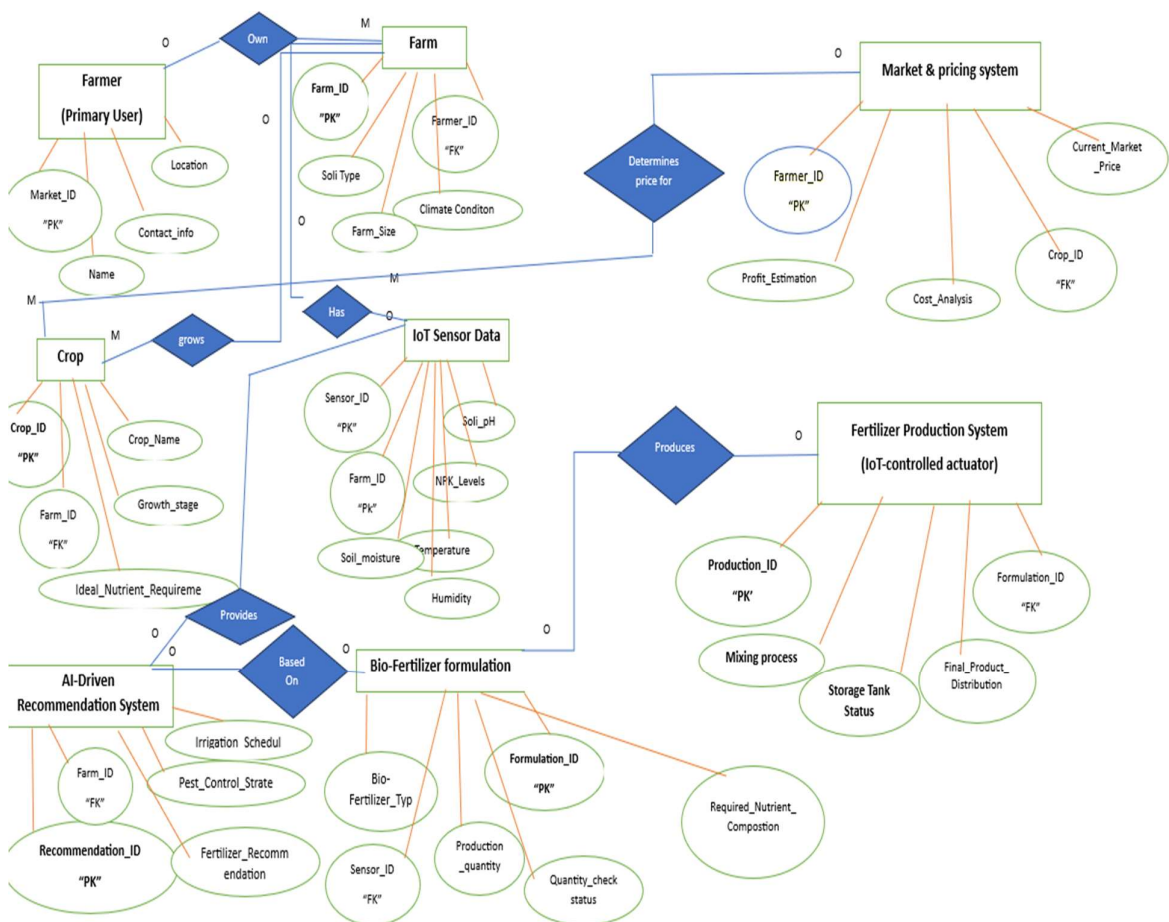


Fig. 3.1 Entity-Relationship Diagram

This image is an Entity-Relationship (ER) diagram that shows how different parts of a smart farming system are connected. It starts with Farmers who own Farms. Each farm has details like its location, soil type, size, and climate condition. Farms grow different Crops, and each crop has information like its name, growth stage, and nutrient needs. Farms are also

equipped with IoT Sensors that collect real-time data such as soil pH, moisture, temperature, and humidity, which helps in better farm management.

Based on the sensor data and crop requirements, an AI-Driven Recommendation System provides suggestions for irrigation schedules, pest control strategies, and fertilizer recommendations. A Bio-Fertilizer Formulation system then prepares the right type and quantity of fertilizer based on the recommendations. Meanwhile, a Market and Pricing System helps farmers by estimating profits and analyzing costs according to current market prices. Finally, a Fertilizer Production System (controlled by IoT actuators) handles the actual production, mixing, and storage of fertilizers, ensuring farmers get the right product for their farms.

3.2 MODULE SPECIFICATION

Farm Data Management Module:

This module is responsible for collecting, storing, and managing all farm data, including soil health, weather data, crop history, and irrigation cycles. IoT sensor data such as soil moisture, temperature, and humidity sensors are collected and stored in this module. It provides functionality to add, edit, and delete farm profiles and synchronizes with the AI Decision Engine for real-time recommendations.

AI-Based Decision Engine Module:

The AI Decision Engine analyzes Farm Data Management Module data to generate prudent suggestions on irrigation, fertilization, and crop rotation. By applying machine learning techniques, it calculates optimal agriculture practice estimates from past patterns, weather patterns, and plant varieties. It is displayed graphically in the Dashboard Module to facilitate easy understanding by farmers and agronomists.

IoT Integration and Monitoring Module:

This module is in charge of real-time communication with the IoT devices deployed on the farm. It receives sensor data on a periodic basis and updates the system to current field conditions. It also performs device status checking and alerts the user in case of any deviation. This module forwards data to the Farm Data Management Module for processing.

Visualization and Dashboard Module:

The Dashboard Module gives a simple visual representation of significant farm statistics and AI advice. The module shows charts, graphs, and alerts so the user can easily assess the state of their farm. Information is fetched from the AI Decision Engine Module and Farm Data Management Module to display actionable details in an understandable format.

User Interface Module:

This module offers a seamless interaction experience for various users such as farmers, agronomists, and system administrators. It is role-based access control enabled and responsive design that can be used across devices. The UI Module also supports easy navigation between modules and allows users to input or query data with less effort

3.3 METHODOLOGY OF THE PROPOSED SYSTEM

Data Source

The data was gathered from various IoT-based agricultural sensors scattered throughout the farm. They consisted of sensors that measured soil moisture, temperature, humidity, light intensity, and water levels. External inputs were also received from weather APIs and crop history databases.

The data was rich and granular but complex and needed processing to yield useful insights for farmers.

Objective

The main goal was to create an intelligent decision support system that streamlines farm data analysis and provides actionable suggestions. The system is designed to maximize crop planning, irrigation schedules, and fertilizer application using data visualization and AI-driven insights.

System Design

- **GUI Creation**

A user-friendly and intuitive interface was created to allow farmers and agronomists to interact with the system. The GUI allows real-time monitoring of data, generation of reports, and management of IoT devices.

- **Input Handling**

Sensor information gets automatically pushed into the system through wireless communication from IoT sensors. The system also supports manual input of data or uploading in CSV format for past information. Input data is validated, normalized, and stored in tabular form for subsequent processing.

- **Processing**

- 1) Data processing is done with Pandas and NumPy to pre-clean and process raw inputs.
- 2) The AI component employs machine learning algorithms to make predictions and suggestions on irrigation timing, crop compatibility, and yield prediction.
- 3) A rules-based engine combines domain-specific thresholds for alerting farmers regarding anomalies like under-irrigation or pest threats.

- **Visualization**

- 1) Soil & Weather Trends: Line graphs show soil moisture, temperature, and humidity changes over time.
- 2) Crop Health Analysis: Bar charts display comparison of varied crop performances through sensor data.
- 3) Decision Insights: Pie charts and gauges present AI-developed suggestions such as ideal frequency of watering, amount of fertilizer, and risk alerts

3.4. DATA FLOW DIAGRAMS (DFDs)

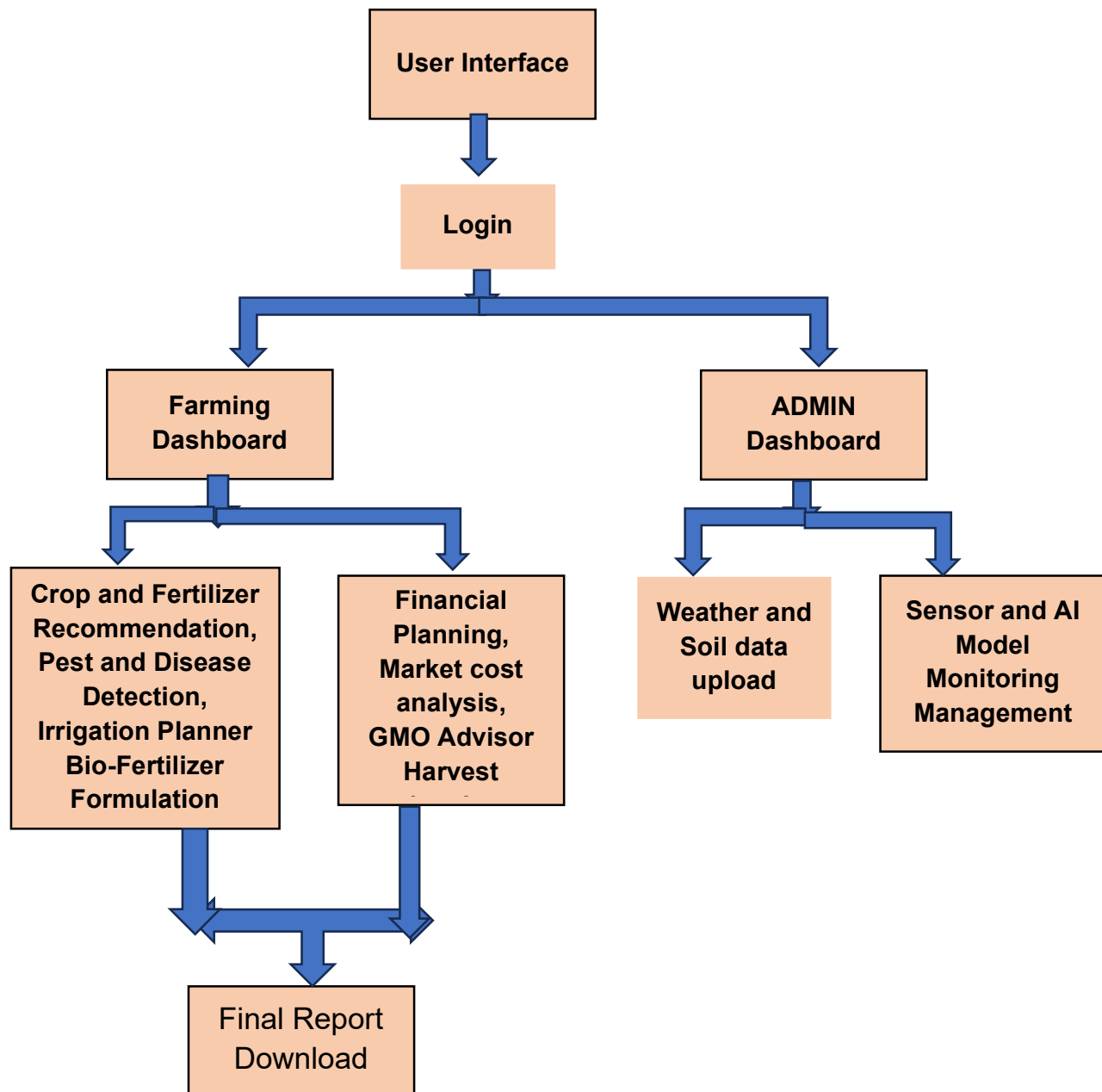


Fig. 3.4 Data Flow Diagrams (DFDs)

The system starts with a user-friendly interface where users log in. After logging in, users are directed either to the Farming Dashboard or the Admin Dashboard. The Farming Dashboard helps farmers with crop and fertilizer recommendations, pest and disease detection, irrigation planning, and bio-fertilizer suggestions. It also supports farmers with financial planning,

market cost analysis, GMO advice, and harvest planning. All the information collected and processed here leads to a final report that the farmer can easily download for future use.

On the other side, the Admin Dashboard is designed for administrators to manage the system's backend. Admins can upload important weather and soil data that the system needs to give accurate recommendations. They also monitor and manage the sensors and AI models to ensure everything runs smoothly. Together, both dashboards work to provide farmers with complete support, helping them make better farming decisions and improve their productivity.

3.5. TABLE SPECIFICATION

The information used to create and test the Farming Decision Support System (FDSS) was gathered from IoT-based sensors installed throughout the agricultural ecosystem, as well as external environmental data sources. The dataset contains a complete record of real-time and historical agricultural parameters, enabling intelligent analysis and informed decision-making. The dataset structure consists of several features related to farm conditions, crop health, and environmental conditions. The main components are described below:

Sensor ID / Location ID - Every record is assigned a unique identifier to a particular sensor or location on the farm. This allows precise tracking and linking of readings with physical locations.

•

Timestamp - Shows the precise time and date on which the data was captured. This facilitates analysis of time-series trends and fluctuations over various periods (e.g., hourly, daily).

•

Environmental and Soil Parameters

- Soil Moisture (%) – Quantifies the amount of water in the soil to define irrigation requirements.
- Soil Temperature (°C) – Useful for root zone condition evaluation.
- Air Temperature (°C) – Affects crop development and disease susceptibility.
- Humidity (%) – Influences plant transpiration and water storage.
- Light Intensity (Lux) – Used for photosynthesis and growth evaluation.
- Rainfall (mm) – Used in water budgeting and irrigation planning.

Crop Metadata

- Crop Type – Indicates the crop type being cultivated (e.g., wheat, rice, tomatoes).
- Growth Stage – Reflects the growth stage of the crop (e.g., seeding, flowering, harvesting).
- Fertilizer Application Rate – Recorded manually or by IoT-enabled applicators.

AI-Predicted Recommendations

- Irrigation Schedule–Proposed watering time and volumes based on real-time information.
- Fertilization Plan–Suggests application of nutrients depending on soil and crop requirements.
- Risk Alerts–Created for pest infestation, water stress, or adverse weather.

Key Factors Employed for Plotting and Analysis

1. Sensor ID / Location ID

Employed to segregate data trends applicable to areas of the farm and identify localized faults or variations.

2. Soil Moisture & Temperature

Essential for visualizing irrigation requirements and identifying drought stress.

3. Crop Performance over Time

Time-series plots of sensor readings assist in determining how environmental factors affect growth during different stages of crops.

4. AI Recommendations vs. Actual Conditions

Visual comparisons are drawn between anticipated suggestions and reality in order to quantify system efficiency and inform future optimization.

3.6. USER INTERFACE DESIGN (SCREENS)

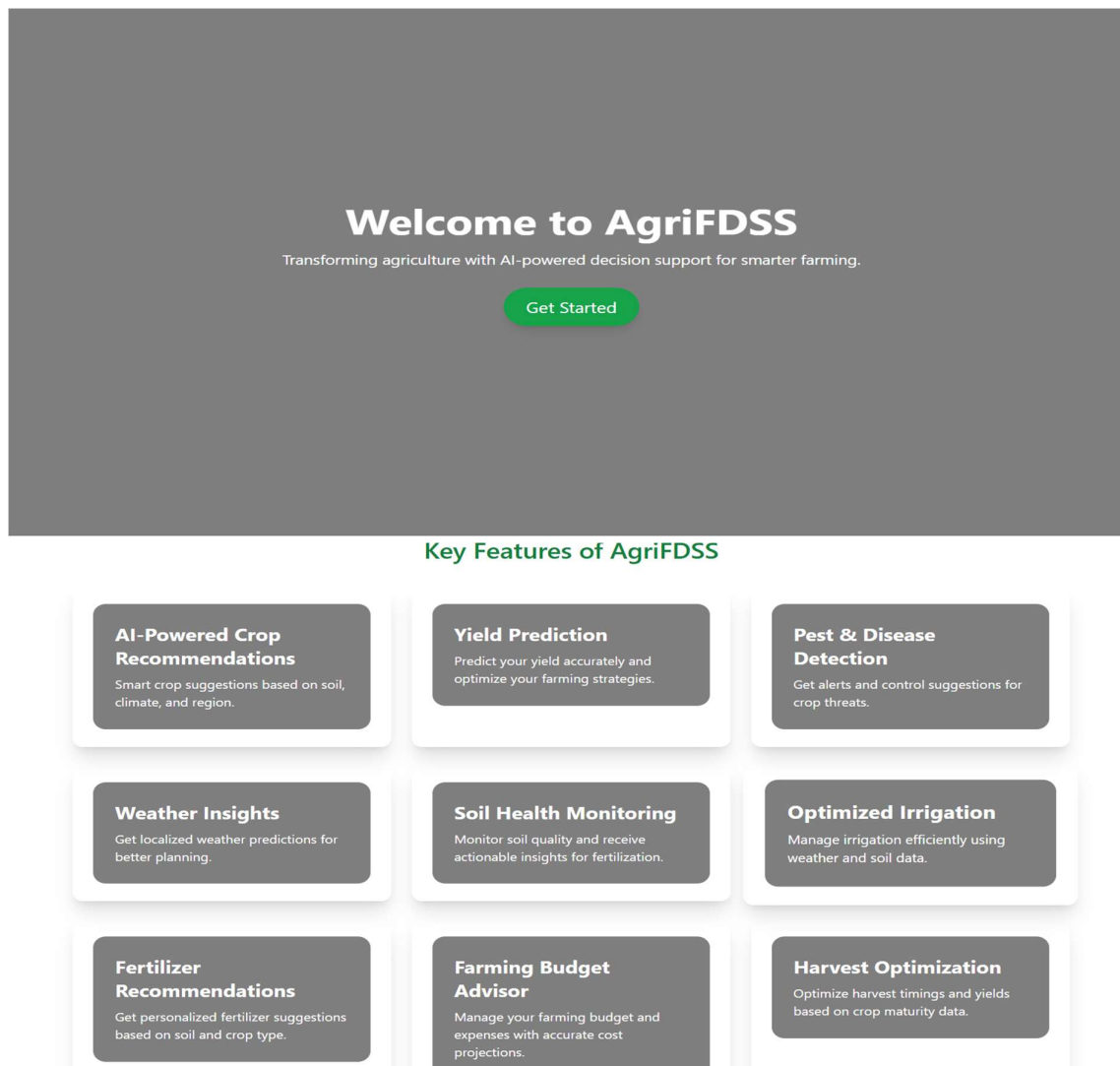
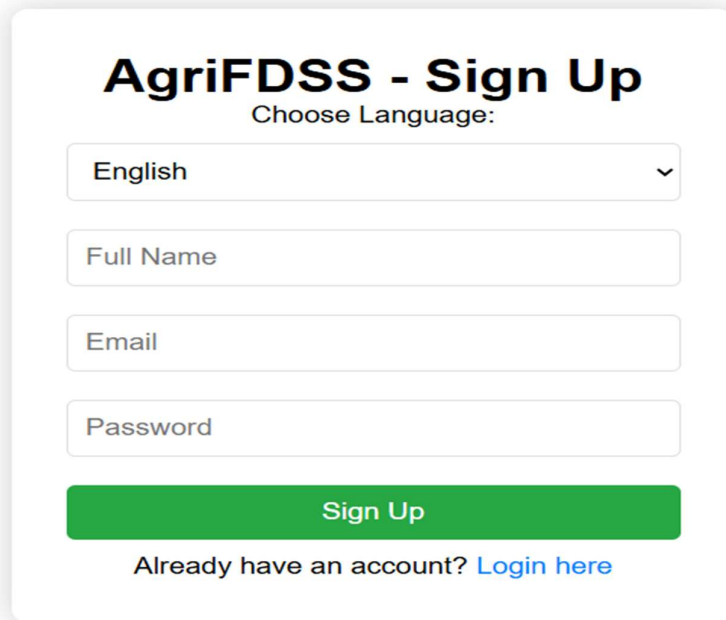


Fig. 3.6.1 Interface

AgriFDSS welcomes users with a focus on transforming agriculture through AI-powered decision support for smarter farming. It offers a range of intelligent features including AI-powered crop recommendations, accurate yield prediction, and pest and disease detection. Farmers can access localized weather insights, monitor soil health, and manage irrigation efficiently using real-time data. The platform also provides personalized fertilizer recommendations, farming budget management, and harvest optimization to maximize productivity. Together, these features empower farmers to make data-driven decisions for sustainable and profitable farming.

The image shows a sign-up form for AgriFDSS. At the top, the title "AgriFDSS - Sign Up" is displayed in a large, bold, black font. Below the title, the text "Choose Language:" is shown. Underneath, there is a dropdown menu with "English" selected and a downward arrow icon. Below the language selection, there are three input fields: "Full Name", "Email", and "Password", each with a light gray border and placeholder text. At the bottom of the form is a prominent green button with the text "Sign Up" in white. Below the button, the text "Already have an account?" is followed by a blue hyperlink labeled "Login here".

AgriFDSS - Sign Up

Choose Language:

English ▼

Full Name

Email

Password

Sign Up

Already have an account? [Login here](#)

Fig. 3.6.2 SignUp Page

The image shows the SignUp page for AgriFDSS (Agriculture Farming Decision Support System). Users are prompted to choose their preferred language from a dropdown menu and then enter their full name, email address, and password to create an account. A green "Sign Up" button is provided for submitting the registration details. Additionally, there is a link at the bottom for users who already have an account to log in.

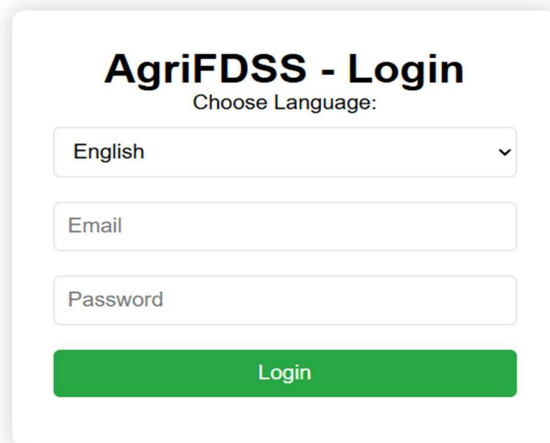
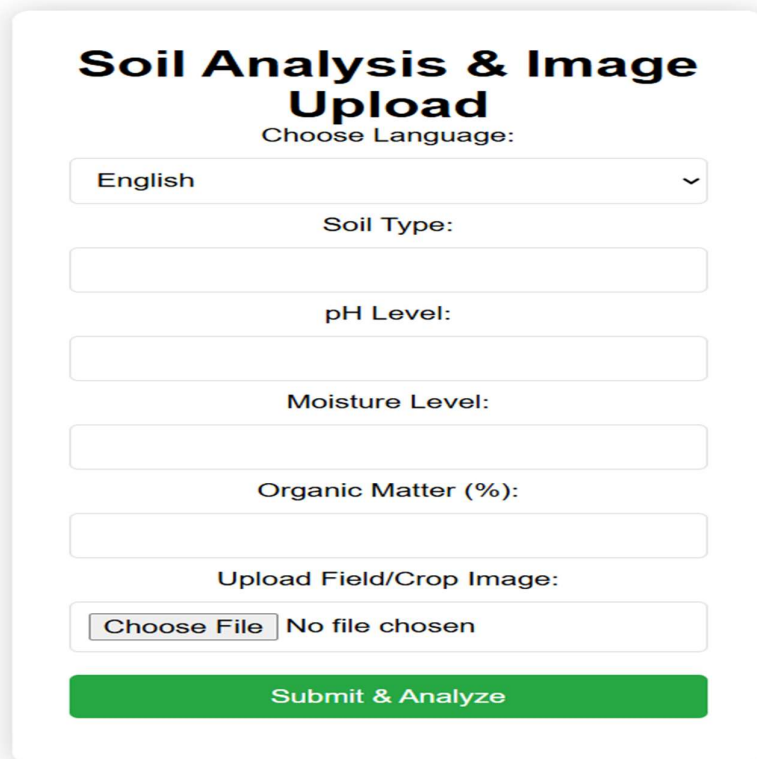
The image shows a login form for AgriFDSS. At the top, the title "AgriFDSS - Login" is displayed in a bold, black font. Below the title, there is a label "Choose Language:" followed by a dropdown menu currently showing "English" with a downward arrow. Underneath the language selection, there are two input fields: one labeled "Email" and another labeled "Password". At the bottom of the form is a prominent green button with the text "Login" in white.

Fig. 3.6.3 Login Page

The AgriFDSS login page offers a simple and secure gateway for users to access the platform. It features a language selection option, along with fields for email and password input. Designed with a clean and minimal interface, it ensures a user-friendly experience. The prominent green "Login" button provides quick access to AgriFDSS's smart farming solutions.

A screenshot of a web form titled "Soil Analysis & Image Upload". The form is centered on a white background with a subtle shadow. It contains a language selection dropdown set to "English", followed by five text input fields for "Soil Type:", "pH Level:", "Moisture Level:", "Organic Matter (%)", and "Upload Field/Crop Image:". The "Upload Field/Crop Image:" field includes a "Choose File" button and the text "No file chosen". At the bottom is a green "Submit & Analyze" button.

Soil Analysis & Image Upload

Choose Language:

English

Soil Type:

pH Level:

Moisture Level:

Organic Matter (%):

Upload Field/Crop Image:

Choose File No file chosen

Submit & Analyze

Fig. 3.6.4 Soil Analysis

The AgriFDSS platform offers a simple login page where users can select their language, enter email and password, and quickly access smart farming solutions. After logging in, users can navigate to the Soil Analysis & Image Upload section. Here, they input details like soil type, pH level, moisture, and organic matter percentage. An option to upload a field or crop image helps in generating detailed analysis and recommendations.

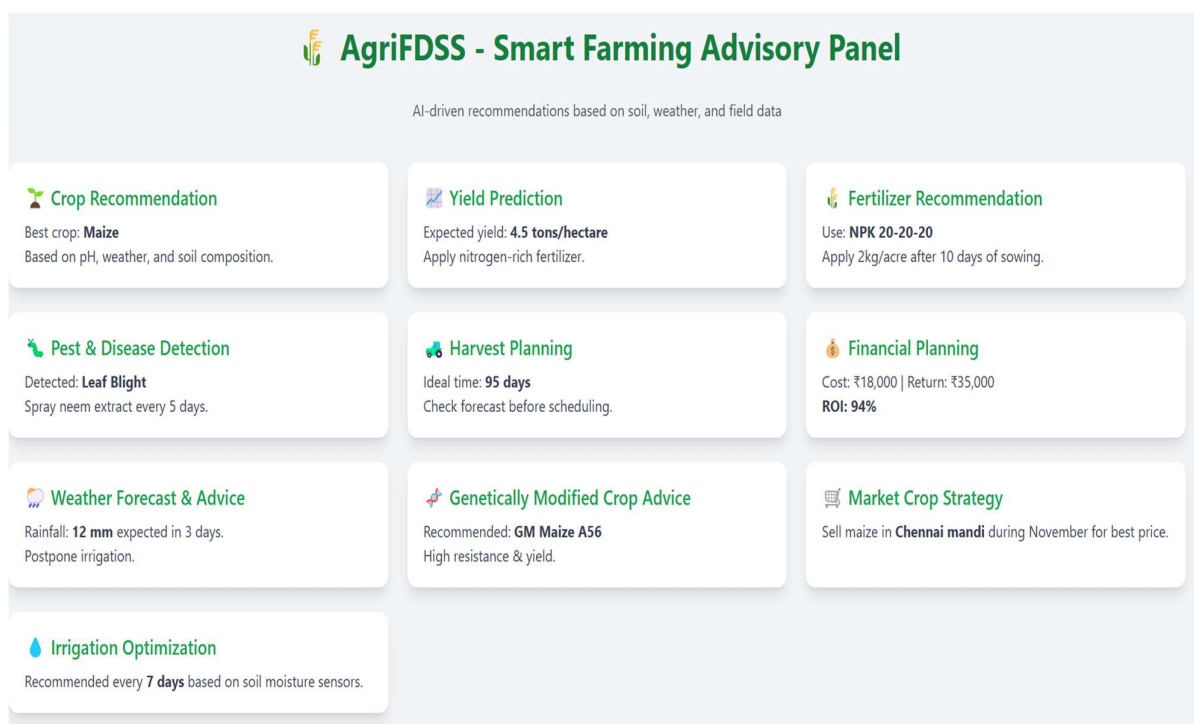
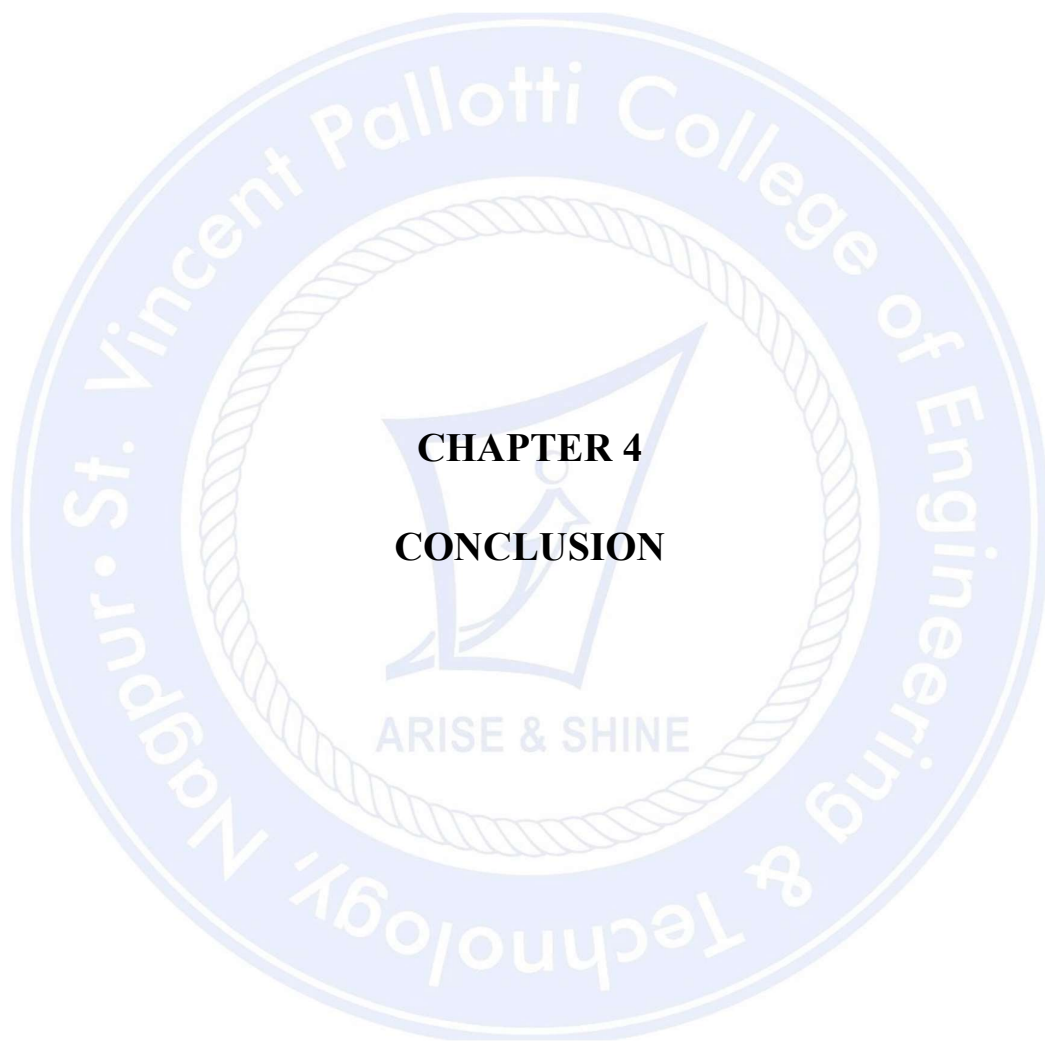


Fig. 3.6.5 Farming Advisory Page

The image shows the AgriFDSS - Smart Farming Advisory Panel, offering AI-driven agricultural recommendations. It provides advice on crop selection, yield prediction, fertilizer usage, pest control, harvest timing, and financial planning based on soil, weather, and field data. Key highlights include recommending maize cultivation, forecasting rainfall to optimize irrigation, and advising on using GM Maize A56 for higher resistance and yield. Financial insights like cost, return, and ROI, along with market strategies, are also given. Overall, it presents a complete decision-support system to enhance farming efficiency and profitability.



CONCLUSION

The development of the AI-Driven and IoT-Integrated Farming Decision Support System is a significant step toward the modernization of agriculture with the use of the convergence of smart analytics and real-time environmental monitoring. By using IoT sensors to collect detailed field data and AI algorithms to make decisions, the system empowers farmers with actionable information to optimize irrigation, fertilization, and crop selection. Not only does this enhance productivity but also resource efficiency and sustainable agriculture.

The system's modular architecture, real-time dashboards, and predictive capabilities represent its capacity to make conventional farming a data-driven practice. With further scalability and incorporation of more sophisticated AI models, the system can be adapted to support multiple crops and environmental parameters, making it suitable for small-scale and large-scale commercial farming practices. Lastly, this project demonstrates how technological innovation can mitigate some of the most serious issues that face agriculture and be part of contributing to the future of smart farming.

4.1 LIMITATIONS

- **Limited Dataset:** The system's accuracy depends on the availability and quality of sensor data, which may be limited in smaller farms or during initial setup.
- **Connectivity Dependency:** Reliable internet or network connectivity is required for real-time IoT data transmission and cloud processing.
- **Scalability Challenges:** The current implementation may require optimization to handle larger farms or multiple crop types simultaneously.
- **Weather API Accuracy:** External weather data sources may not always provide hyper-local predictions, affecting recommendation accuracy.
- **Maintenance of IoT Devices:** Sensors require regular calibration and maintenance to ensure accurate data collection.

4.2. FUTURE SCOPE

- **Advanced Predictive Analytics:**
Integrate machine learning algorithms to predict crop yields, pest infestations, and soil degradation trends for anticipatory decision-making.
- **Mobile Application:**
Create a separate mobile application to increase system access for farmers and agronomists to monitor and control in real-time from remote locations.
- **Real-time Alerts and Notifications:**
Incorporate immediate alerts for urgent conditions like low soil moisture, weather alerts, or equipment faults to enhance responsiveness.
- **Integration with External Platforms:**
Facilitate smooth integration with weather forecasting services, government agricultural databases, and farm management platforms for enhanced functionality.
- **Improved Data Security:**
Secure the system with robust security features such as end-to-end encryption, multi-factor authentication, and access control to safeguard sensitive farm data.
- **Multilingual Interface:**
Integrate multilingual support to serve farmers from various regions and linguistic backgrounds, promoting broader usability and adoption.
- **Advanced Visualization Tools:**
Add more dynamic, interactive, and customizable data visualizations to dashboard features for enhanced analysis and decision-making support.
- **Cloud-Based Deployment:**
Move to a cloud-based infrastructure to enhance scalability, provide round-the-clock data backup, and enable access from anywhere or on any device.

The FDSS has the potential to transform current farm practices. Addressing the limitations and adding these improvements, it can become an intelligent, comprehensive, and valuable tool for precision agriculture and green farming.

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**PROJECT GUIDE
AND
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